Neuromorphic Bootcamp

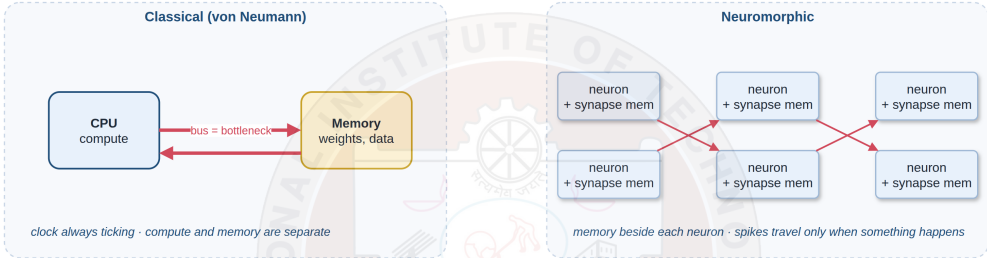
From Spikes to Silicon: the basics of neuromorphic computing, hands-on in Verilog



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This workshop is conducted as a national service under the PMRF external teaching assistantship.

Classical vs neuromorphic computing



Classical (von Neumann)

- CPU and memory are separate boxes
- data moves over a bus every step: the bottleneck
- a global clock drives everything, all the time

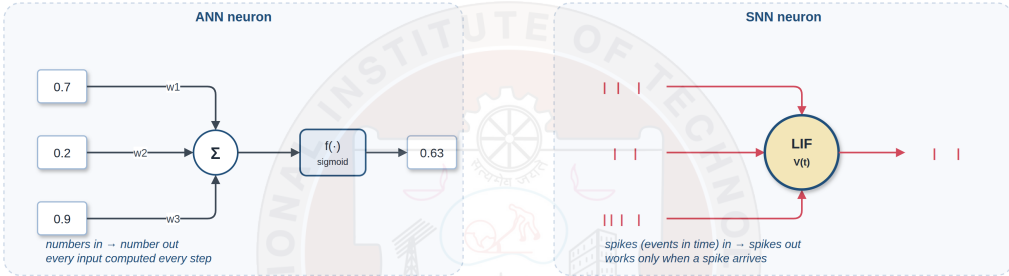
Neuromorphic

- many small neuron cores, memory beside each one
- communicate with spikes (short events)
- compute only when a spike arrives (event-driven, low power)

Examples: IBM TrueNorth, Intel Loihi, SpiNNaker, BrainScaleS

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ANN vs SNN: what changes



ANN (artificial neural network)

- real numbers in and out
- no notion of time inside the neuron
- every neuron computes every step

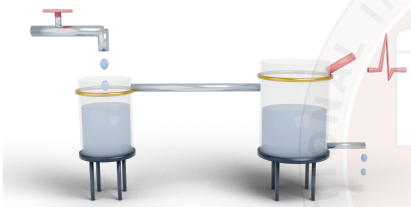
SNN (spiking neural network)

- spikes (0/1 events) in time
- the neuron has a state V(t) that evolves
- information is in timing and rate of spikes

SNN = ANN + time + events. This workshop builds the SNN parts in Verilog.

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One analogy for the whole workshop



in the picture		in the neuron
tank water level	→	**membrane voltage V**
inflow pipe	→	**input current I**
gold mark → dump	→	**threshold → spike, reset**
drain at the bottom	→	**leak (τ)**
pipe between tanks	→	**synapse**
pipe width	→	**synaptic weight w**
pipes widen / narrow	→	**learning (STDP)**

Every model changes exactly one part of this picture.

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How the hardware is built

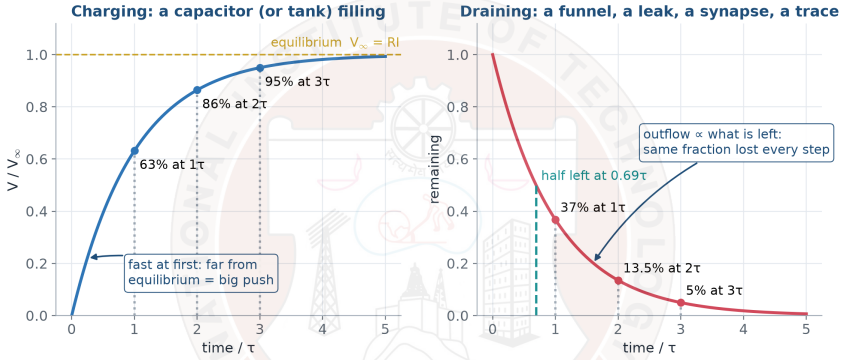
- one clock edge = one Euler step of the differential equation
- fixed point Q8: value $\times$ 256 (20 mV $\rightarrow$ 5120), 32-bit signed
- $dt/\tau = 2^{-k}$, so dividing by τ is a right shift (no divider)
- every testbench writes a CSV, every plot on these slides is from that CSV

Icarus Verilog | Vivado 2025.2 xsim | Vivado synthesis (Artix-7) | Python / matplotlib

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The one curve you will see everywhere: the exponential

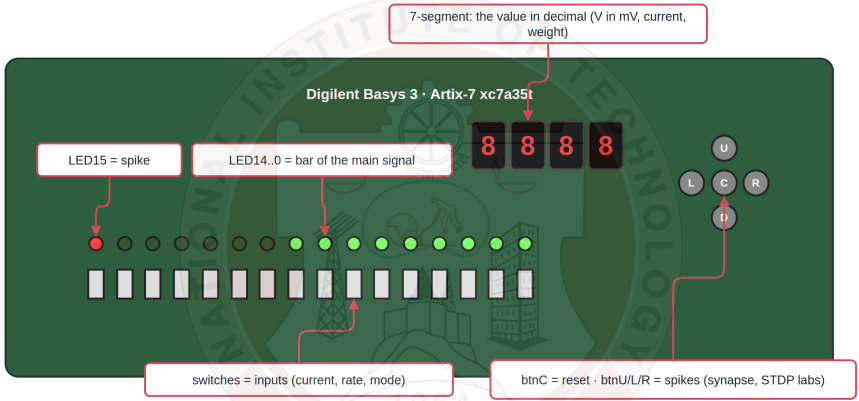
$dX/dt = -(X - X_\infty)/\tau \quad \rightarrow \quad$ the rate is proportional to the distance from equilibrium



- a leaky membrane, a synaptic funnel and an STDP trace all obey the same law: the rate of change is proportional to how far the system is from equilibrium
- so every step removes the same fraction: after one time constant τ, 63% of the way is done (charging) or 37% is left (draining)
- in hardware this is one shift: $X \leftarrow X - (X \gg k)$, with $\tau = 2^k$ steps

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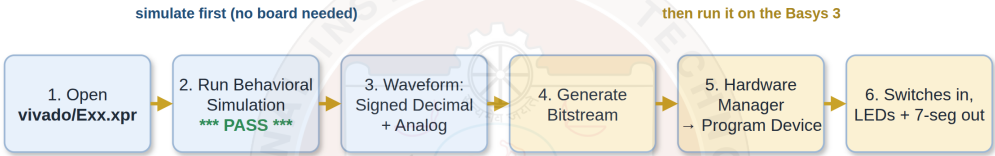
Your lab kit: the Basys 3 board



- the same controls in every lab, so you learn them once
- each model step is slowed down to 20-1000 steps per second, so you can watch the neuron think

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How every lab runs



- every experiment is a ready Vivado project: experiments/Exx_name/vivado/Exx_name.xpr
- the testbench checks itself: the Tcl console prints each value and *** PASS ***
- the bitstream is already generated; after you change the RTL, click Generate Bitstream again
- no board? Everything up to step 3 works in simulation only

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The eleven labs

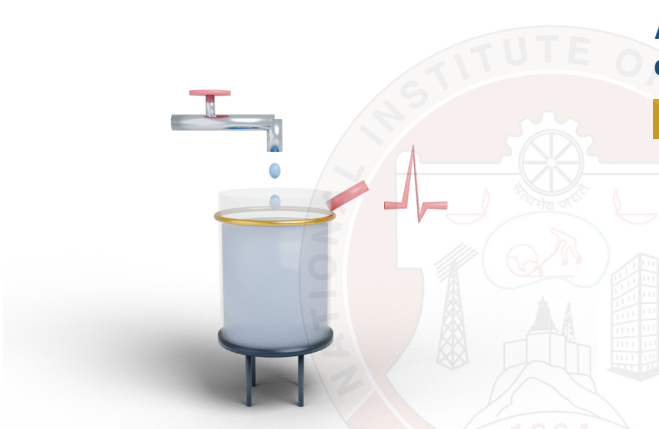
folder	model	one step on the board
`E01_if_neuron`	Integrate-and-Fire neuron	1/50 s
`E02_lif_neuron`	Leaky Integrate-and-Fire neuron	1/50 s
`E03_adaptive_lif`	Adaptive LIF neuron	1/50 s
`E04_qif_neuron`	Quadratic Integrate-and-Fire neuron	1/50 s
`E05_izhikevich`	Izhikevich neuron: four cell types	1/200 s
`E06_delta_synapse`	Delta synapse	1/20 s
`E07_exp_synapse`	Exponential synapse	1/20 s
`E08_alpha_synapse`	Alpha synapse	1/20 s
`E09_tsodyks_markram`	Tsodyks-Markram dynamic synapse	1/50 s
`E10_stdp_synapse`	STDP synapse: learning from spike timing	1/25 s
`E11_stdp_network`	STDP network: unsupervised learning	1/1000 s

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Neurons

N1 Integrate-and-Fire (IF)
N2 Leaky Integrate-and-Fire (LIF)
N3 Adaptive LIF
N4 Quadratic Integrate-and-Fire (QIF)
N5 Izhikevich neuron

N1 IF neuron



Analogy: A tank that fills and dumps

NEW: the starting point

- tank water level = membrane voltage V; inflow pipe = input current I
- the gold mark is the threshold: when the level reaches it, the tank dumps all its water at once
- that dump is the spike; the tank starts again from empty (reset)
- the tank is sealed: water never escapes, so even a slow trickle fills it eventually

Lapicque 1907 ▪ lab: `experiments/E01_if_neuron`

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N1 IF neuron: the equation and its physics

MODEL

$$\underbrace{C\frac{dV}{dt}}_{\text{charging}} = \underbrace{I(t)}_{\text{input}}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$V[n{+}1] = \underbrace{V[n]}_{\text{memory}} + \underbrace{I[n]}_{\text{add input}}$$

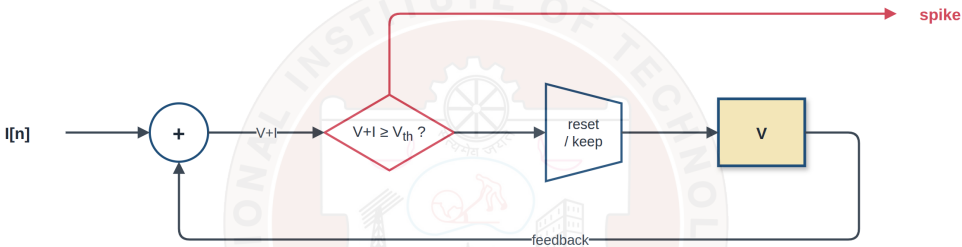
if $V \geq V_{th}$: **spike**, $V \leftarrow V_{reset}$

The physics: A capacitor charged by a constant current: C dV/dt = I, so the voltage rises in a straight line (Q = C·V). Nothing leaks, so any current eventually reaches the threshold.

- the voltage is the running sum of the input
- threshold + reset turn a sum into spikes

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N1 IF neuron: hardware



1 adder · 1 comparator · 1 register — the simplest neuron

- one adder, one comparator, one register
- the register holds V between clock edges

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N1 IF neuron: Verilog

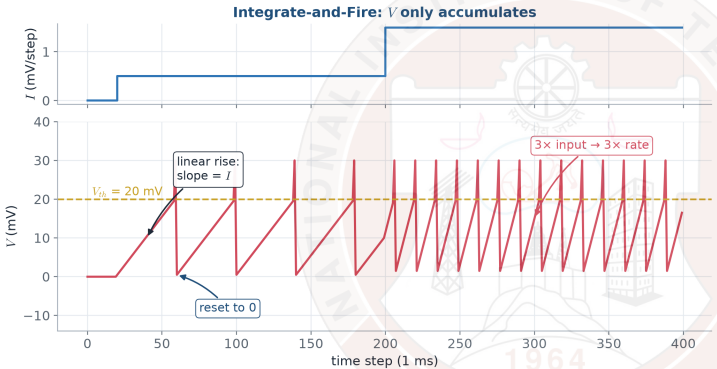
```
    wire signed [31:0] v_next = v + i_in;
    always @(posedge clk) begin
        if (rst) begin v <= VRESET; spike <= 1'b0; end
        else if (en) begin
            if (v_next >= VTH) begin v <= VRESET; spike <= 1'b1; end
            else               begin v <= v_next; spike <= 1'b0; end
        end else spike <= 1'b0;
    end
endmodule
```

rtl/neurons/if_neuron.v

- v_next = v + i_in is the whole model
- compare with VTH, then reset or keep
- spike is a 1-cycle pulse

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N1 IF neuron: neural dynamics



How to read it: x-axis: time steps (1 step = 1 ms). Top: the input. Bottom: V ramps in straight lines, jumps to 0 at each spike.

- linear ramps: slope equals the input
- 3× more input gives 3× more spikes
- no leak, so even tiny input fires (unrealistic)

In the lab you should see: spike every 40 steps at 0.5 mV/step, every 14 steps at 1.5 mV/step (18 spikes in total).

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N1 IF neuron: lab E01 on the Basys 3

Folder: `experiments/E01_if_neuron/`
Open: `vivado/E01_if_neuron.xpr`

control	what it does
`sw[7:0]`	input current I (Q8: 128 = 0.5 mV/step)
`btnC`	reset
`LED14..0`	membrane voltage V (bar, full = threshold 20 mV)
`LED15`	spike
`7-segment`	V in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw = 0000 0001: very slow ramp, still fires (no leak!)
- sw = 1000 0000: about one spike per 0.8 s
- double the current: the spike rate doubles

Simulation must print PASS with:

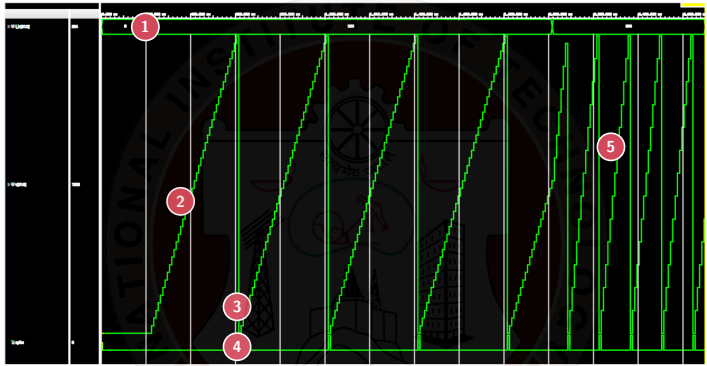
```
spikes    18
```

Try it: Halve VTH in if_neuron.v. Predict the new spike count, then simulate.

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N1 IF neuron: Vivado simulation

input current, V and spikes



1 input switches on 2 V rises in a straight line 3 threshold → spike, V back to 0 4 spike output

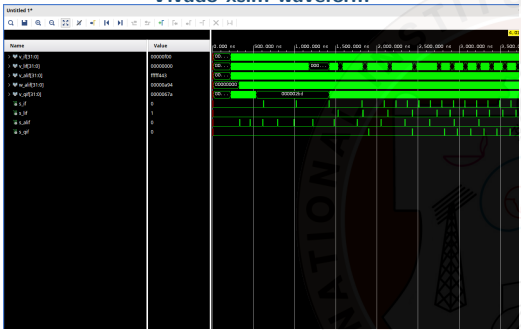
5 3× input → 3× steeper ramps

How to read it: Top row: the input current (signed decimal). Middle row: V drawn as an analog curve. Every ramp is straight because nothing leaks; the bigger the input, the steeper the ramp and the more spikes.

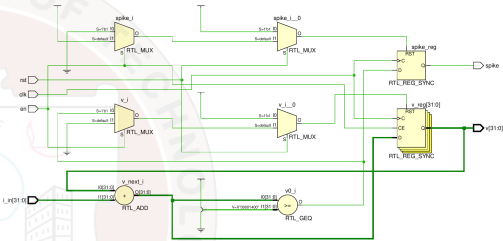
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N1 IF neuron: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic

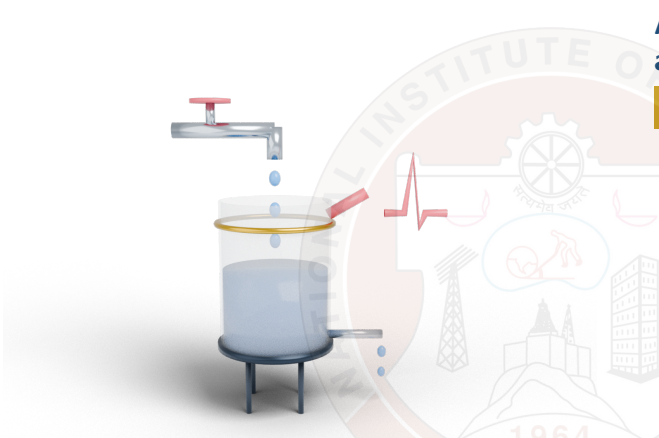


LUT	FF	DSP	F_{max}
96	33	0	161 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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N2 LIF neuron



Analogy: The same tank with a drain

NEW: a drain (leak)

- a small drain at the bottom: the fuller the tank, the faster it drains (outflow $\propto$ level)
- a weak inflow is balanced by the drain: the level settles below the mark, no spike
- only an inflow stronger than the drain can reach the mark: a threshold current
- after each dump the inlet is locked for a moment: the refractory period

Gerstner & Kistler 2002 ▪ lab: `experiments/E02_lif_neuron`

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N2 LIF neuron: the equation and its physics

MODEL

$$\underbrace{\tau \frac{dV}{dt}}_{\tau = RC} = \underbrace{-(V - V_{rest})}_{\text{leak back to rest}} + \underbrace{R\,I(t)}_{\text{drive}}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$V[n{+}1] = V[n] + \underbrace{\big(RI - (V - V_{rest})\big) \gg k}_{\times\, dt/\tau = 2^{-k}\text{: a shift}}$$

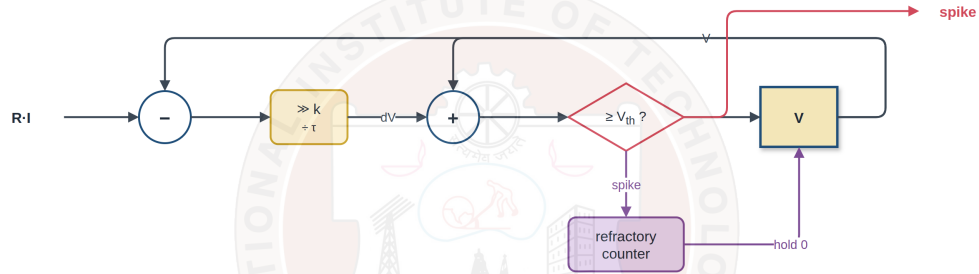
if $V \geq V_{th}$: **spike**, $V \leftarrow V_{reset}$, hold for T_{ref} steps (fires only if $RI > V_{th}$)

The physics: An RC circuit: the membrane capacitor charges through the input and leaks through a resistor. The leak current is proportional to V (Ohm's law), so V approaches R·I exponentially with time constant $\tau = RC$. If R·I is below the threshold, it never fires.

- the leak pulls V back to rest with time constant τ
- steady state $V\infty = RI$: fires only if $RI > Vth$

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N2 LIF neuron: hardware



leak = one subtraction + one shift ($dt/\tau = 2^{-k}$): no multiplier

- the leak costs one subtract and one shift
- a small counter implements the refractory period

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N2 LIF neuron: Verilog

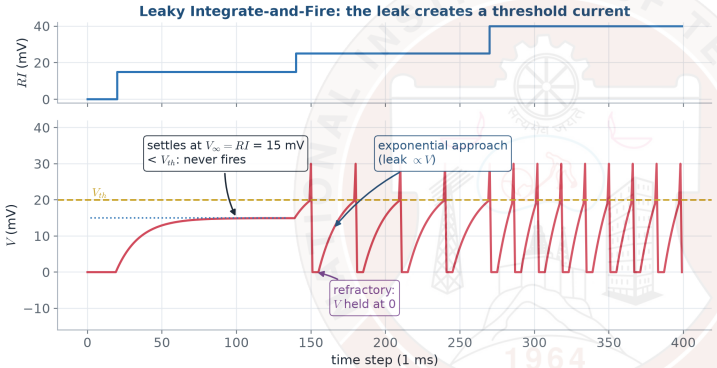
```
    reg [7:0] ref_cnt;
    wire signed [31:0] dv     = (i_in - (v - VREST)) >>> TAU_SHIFT;
    wire signed [31:0] v_next = v + dv;
    always @(posedge clk) begin
        if (rst) begin v <= VREST; spike <= 1'b0; ref_cnt <= 0; end
        else if (en) begin
            spike <= 1'b0;
            if (ref_cnt != 0) begin ref_cnt <= ref_cnt - 1'b1; v <= VRESET; end
            else if (v_next >= VTH) begin
                v <= VRESET; spike <= 1'b1; ref_cnt <= T_REF[7:0];
            end else v <= v_next;
        end else spike <= 1'b0;
    end
endmodule
```

rtl/neurons/lif_neuron.v

- dv = (i_in – (v – VREST)) »> TAU_SHIFT
- ref_cnt holds V at reset for T_REF steps
- TAU_SHIFT = 4 means τ = 16 steps

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N2 LIF neuron: neural dynamics



How to read it: Watch the shape of each ramp: curved, because the leak grows with V. At 15 mV the curve flattens before the threshold.

- 15 mV input settles below threshold: silent
- curved (exponential) charging
- flat segments after each spike = refractory

In the lab you should see: 15 mV: V settles near 15 mV, 0 spikes; 25 mV: regular spikes; 40 mV: faster spikes.

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N2 LIF neuron: lab E02 on the Basys 3

Folder: `experiments/E02_lif_neuron/`
Open: `vivado/E02_lif_neuron.xpr`

control	what it does
`sw[5:0]`	R·I in mV (0-63)
`btnC`	reset
`LED14..0`	V (bar, full = 20 mV threshold)
`LED15`	spike
`7-segment`	V in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw = 15: the bar stops below the top, no spikes (below rheobase)
- sw = 21: slow firing just above threshold
- sw = 40: fast firing, with a pause after each spike (refractory)

Simulation must print PASS with:

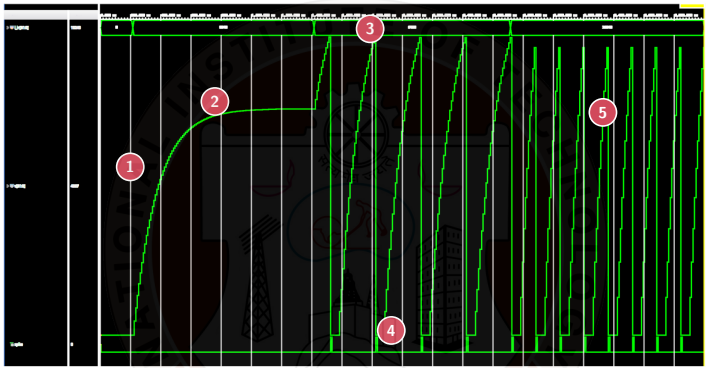
```
spikes_at_15mV    0
vmax_at_15mV_Q8   3825
spikes_at_25mV    4
spikes_at_40mV    9
```

Try it: Change TAU_SHIFT from 4 to 2 (shorter τ). What happens to the charging curve?

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N2 LIF neuron: Vivado simulation

below and above threshold



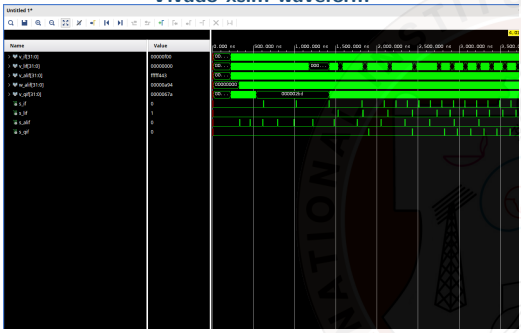
1 curved rise: exponential (RC charging) 2 15 mV: settles below threshold, no spike 3 25 mV: crosses threshold 4 flat after reset: refractory 5 40 mV: faster firing

How to read it: Compare with E01: every ramp now bends over, because the leak grows as V grows. With 15 mV the leak balances the input before the threshold: the neuron never fires.

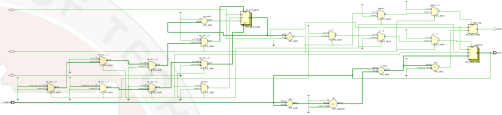
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N2 LIF neuron: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



LUT	FF	DSP	F_{max}
142	41	0	132 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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N3 Adaptive LIF



Analogy: The drain opens wider after every dump

NEW: an extra, slowly closing drain

- each dump opens a second drain valve a little further (+B)
- the valve closes again only slowly ($\tau_w \gg \tau_v$)
- with the same inflow, the tank now fills more slowly after each spike
- so the gaps between spikes grow: spike-frequency adaptation

Brette & Gerstner 2005 ▪ lab: `experiments/E03_adaptive_lif`

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N3 Adaptive LIF: the equation and its physics

MODEL

$$\tau_v \frac{dV}{dt} = -(V - V_{rest}) + RI \underbrace{- w}_{\text{adaptation current}}, \qquad \underbrace{\tau_w \frac{dw}{dt} = -w}_{\text{slow decay, } \tau_w \gg \tau_v}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$V \mathrel{+}= (RI - w - V) \gg k_v, \qquad w \mathrel{-}= w \gg k_w$$

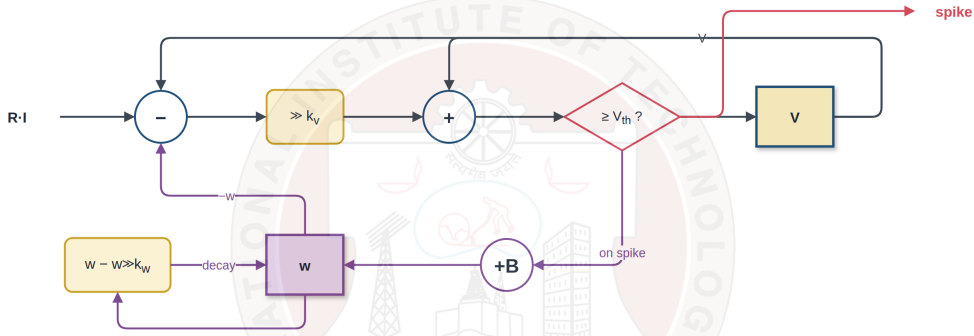
on spike: $V \leftarrow V_{reset}$, $w \leftarrow w + B$ (Brette & Gerstner 2005, linear form)

The physics: Two capacitors with two very different time constants: the membrane (fast, τ_v) and an adaptation current w (slow, τ_w). Each spike adds charge to w; w drains slowly and opposes the input, so the neuron 'gets tired'.

- a second state w subtracts from the drive
- w jumps on each spike, decays slowly

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N3 Adaptive LIF: hardware



NEW: second register w — jumps by B on each spike, decays slowly, subtracts from the drive

- a second register w with its own leak
- the spike signal drives the +B adder

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N3 Adaptive LIF: Verilog

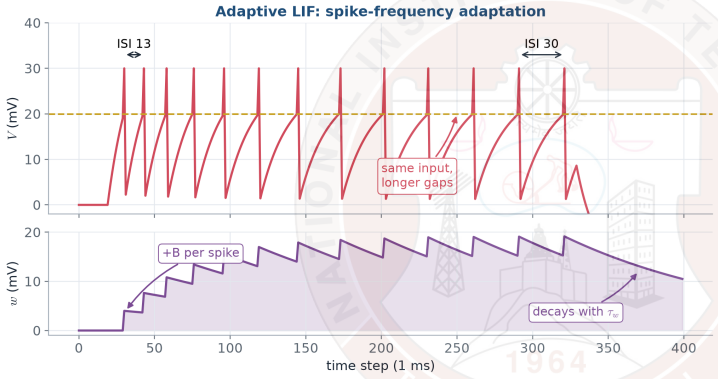
```
    wire signed [31:0] v_next = v + ((i_in - w - (v - VREST)) >>> TAU_V);
    wire signed [31:0] w_dec  = w - (w >>> TAU_W);
    always @(posedge clk) begin
        if (rst) begin v <= VREST; w <= 0; spike <= 1'b0; end
        else if (en) begin
            if (v_next >= VTH) begin
                v <= VRESET; w <= w_dec + B; spike <= 1'b1;
            end else begin
                v <= v_next; w <= w_dec; spike <= 1'b0;
            end
        end else spike <= 1'b0;
    end
endmodule
```

rtl/neurons/alif_neuron.v

- w_dec = w − (w »> TAU_W)
- on spike: w <= w_dec + B
- v uses (i_in − w − v)

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N3 Adaptive LIF: neural dynamics



How to read it: Top: V with the gaps between spikes marked. Bottom: w, the 'tiredness', rising at each spike and draining between.

- constant input, yet the inter-spike interval grows 13 → 30 steps
- w builds up spike by spike
- after the input stops, w decays

In the lab you should see: the inter-spike interval grows from the first spike pair to the last (spike-frequency adaptation).

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N3 Adaptive LIF: lab E03 on the Basys 3

Folder: `experiments/E03_adaptive_lif/`
Open: `vivado/E03_adaptive_lif.xpr`

control	what it does
`sw[5:0]`	R·I in mV
`sw[15]`	7-segment shows V (0) or adaptation w (1)
`btnC`	reset
`LED14..0`	V (bar)
`LED15`	spike
`7-segment`	V or w in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw = 40: watch the LED15 flashes slow down after the first few
- flip sw[15]: w climbs with every spike, then decays when the input stops

Simulation must print PASS with:

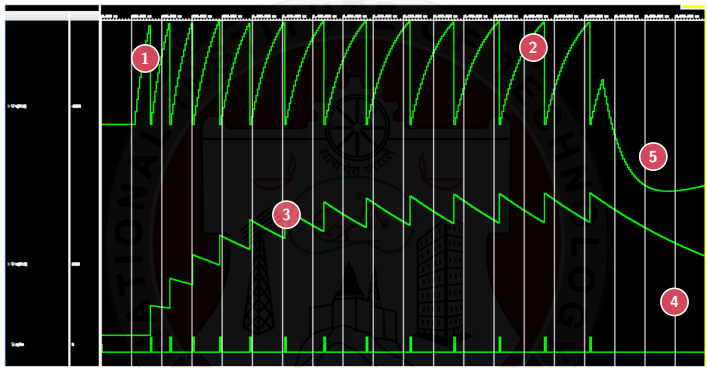
`spikes`	13
`first_ISI_steps`	13
`last_ISI_steps`	30

Try it: Set B = 0. Which experiment's behaviour do you get back?

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N3 Adaptive LIF: Vivado simulation

V and the adaptation current w



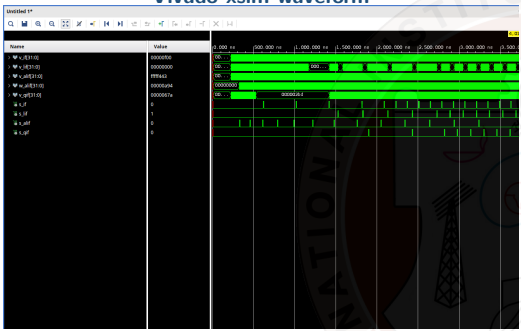
1 first spikes close together 2 same input, wider gaps 3 w jumps at every spike 4 w drains slowly after the input stops 5 input off: V relaxes

How to read it: Top: V. Middle: w, the 'tiredness'. Each spike adds to w and w subtracts from the input, so each next ramp is slower: spike-frequency adaptation.

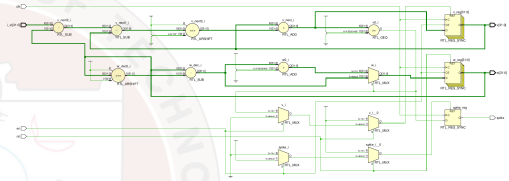
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N3 Adaptive LIF: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic

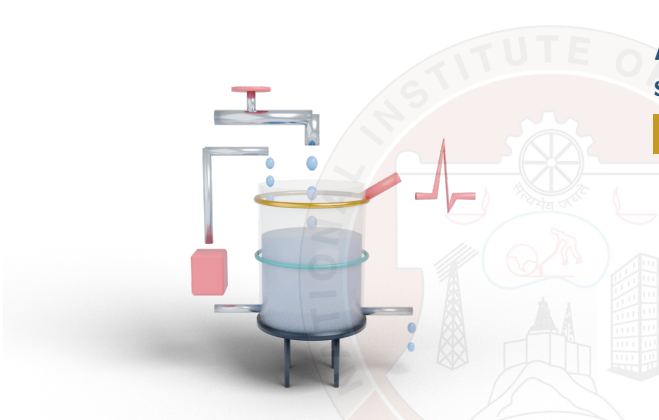


LUT	FF	DSP	F_{max}
206	65	0	112 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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N4 QIF neuron



Analogy: A pump that switches on above a level

NEW: positive feedback

- a pump feeds water back into the tank, faster the higher the level
- below the teal mark Vc the drain wins: the level falls back to rest
- above Vc the pump wins and the level races up by itself: the spike
- the spike onset is now part of the physics, not only a fixed mark

Ermentrout 1996; Latham et al. 2000 ▪ lab: `experiments/E04_qif_neuron`

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N4 QIF neuron: the equation and its physics

MODEL

$$\tau \frac{dV}{dt} = \underbrace{a_0 (V - V_{rest})(V - V_c)}_{\text{quadratic: self-accelerating above } V_c} + RI$$

HARDWARE (ONE CLOCK = ONE STEP)

$$V \mathrel{+}= \Big(\underbrace{\big(V(V - V_c)\big) \gg (8 + a)}_{\text{one multiplier}} + RI \Big) \gg k$$

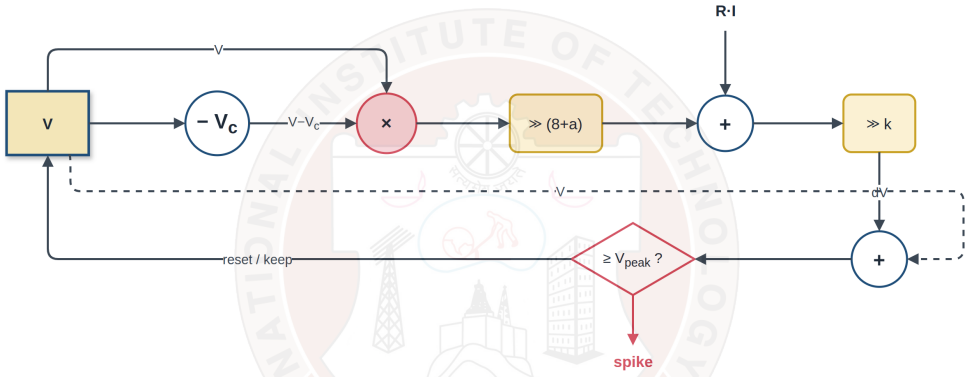
if $V \geq V_{peak}$: **spike**, $V \leftarrow V_{reset}$ rheobase $RI^* = a_0 V_c^2/4$

The physics: Positive feedback: above the critical voltage Vc the term (V–Vc)·V is positive and grows with V, so the voltage accelerates itself, like a ball rolling over a hilltop. The spike onset comes from the equation, not from a fixed rule.

- the quadratic term is negative below Vc, positive above
- rheobase a0·$Vc^2/4$ = 6.25 mV

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N4 QIF neuron: hardware



NEW: one multiplier for $V \cdot (V - V_c)$ → a DSP slice on the FPGA

- NEW: a multiplier for V·(V – Vc)
- on an FPGA this maps to DSP slices

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N4 QIF neuron: Verilog

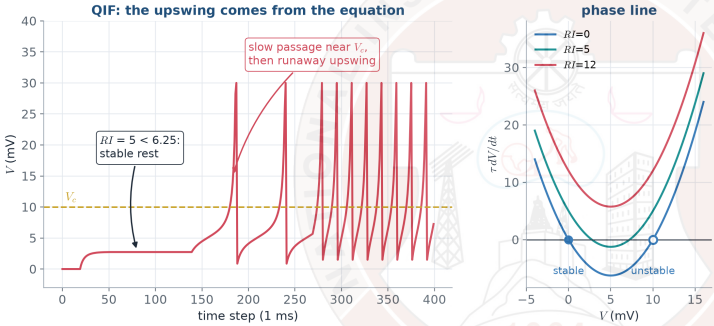
```
    wire signed [63:0] prod   = v * (v - VC);              // Q16
    wire signed [63:0] quad   = prod >>> (8 + A_SHIFT);    // back to Q8, times a0
    wire signed [63:0] dv     = (quad + i_in) >>> TAU_SHIFT;
    wire signed [31:0] v_next = v + dv;
    always @(posedge clk) begin
        if (rst) begin v <= VRESET; spike <= 1'b0; end
        else if (en) begin
            if (v_next >= VPEAK) begin v <= VRESET; spike <= 1'b1; end
            else                 begin v <= v_next; spike <= 1'b0; end
        end else spike <= 1'b0;
    end
endmodule
```

rtl/neurons/qif_neuron.v

- prod = v * (v – VC) in 64 bits
- shift back to Q8 and scale by a0
- VPEAK replaces the threshold test

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N4 QIF neuron: neural dynamics



How to read it: Left: V over time. Right: the phase line dV/dt against V; where it crosses zero the neuron can rest.

- 5 mV: stable rest (two fixed points on the phase line)
- 7 mV: slow passage near Vc, then a runaway upswing
- 12 mV: no fixed point, regular firing

In the lab you should see: 5 mV: silent; 7 mV: slow spikes with a long creep before each; 12 mV: regular firing.

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N4 QIF neuron: lab E04 on the Basys 3

Folder: `experiments/E04_qif_neuron/`
Open: `vivado/E04_qif_neuron.xpr`

control	what it does
`sw[4:0]`	R·I in mV (0-31)
`btnC`	reset
`LED14..0`	V (bar, full = 30 mV peak)
`LED15`	spike
`7-segment`	V in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw = 6: V rests (below rheobase 6.25 mV)
- sw = 7: V creeps up slowly, then shoots up: the self-accelerating onset
- sw = 12: regular firing

Simulation must print PASS with:

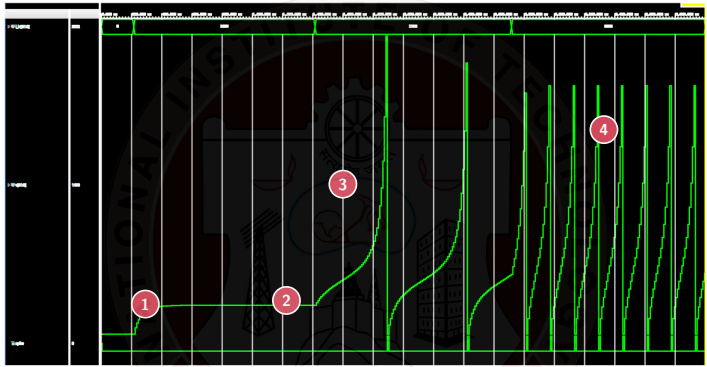
```
spikes_at_5mV    0
spikes_at_7mV    2
spikes_at_12mV   8
```

Try it: Find the exact switch value where firing starts and compare with $a0 \cdot Vc^2/4$.

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N4 QIF neuron: Vivado simulation

rest, slow passage and runaway



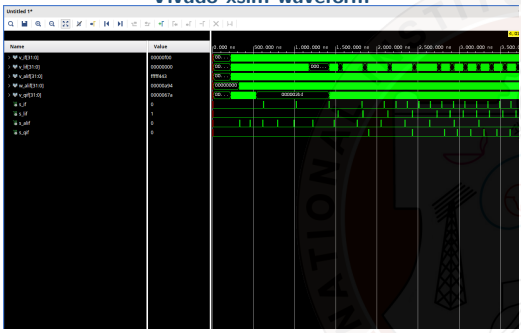
1 5 mV: V rests (stable point) 2 7 mV: slow creep near Vc 3 then a runaway upswing 4 12 mV: regular firing

How to read it: The upswing is not a straight line or a bending curve: it accelerates by itself once V passes Vc, because the quadratic term feeds itself (positive feedback).

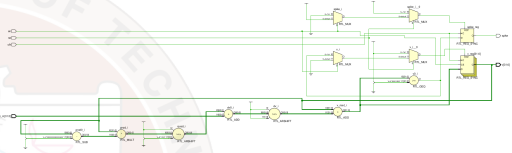
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N4 QIF neuron: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



LUT	FF	DSP	F_{max}
182	33	4	57 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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N5 Izhikevich neuron



Analogy: A main tank with a slow partner tank

NEW: a second, slow variable

- main tank = v (fast); a small partner tank = u (slow recovery)
- the partner fills when the main tank is high, and it drains the main tank
- four knobs: a = partner speed, b = coupling, c = reset level, d = partner jump per spike
- the same tanks with different knob settings give regular, bursting, chattering, fast-spiking

Izhikevich 2003 ▪ lab: `experiments/E05_izhikevich`

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N5 Izhikevich neuron: the equation and its physics

MODEL

$$\frac{dv}{dt} = \underbrace{0.04v^2 + 5v + 140}_{\text{fitted to cortex data}} - \underbrace{u}_{\text{recovery}} + \underbrace{I}_{\text{input}}$$

$$\frac{du}{dt} = \underbrace{a\,(b\,v - u)}_{a\text{: speed, } b\text{: sensitivity}}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$v \mathrel{+}= \big((41\,v^2) \gg 18 + 5v + 140 - u + I\big) \gg 1, \qquad u \mathrel{+}= \big(a(bv - u)\big) \gg 1$$

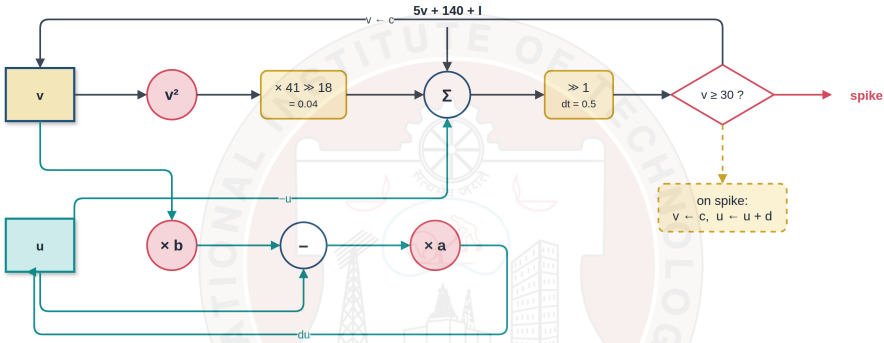
if $v \geq 30$ mV: **spike**, $v \leftarrow c$, $u \leftarrow u + d$ (Izhikevich 2003)

The physics: Two coupled variables: a fast voltage v with a quadratic (self-accelerating) term and a slow recovery u that pulls v back down. Different speeds a and reset values c, d give different rhythms, like springs of different stiffness.

- $0.04v^2 + 5v + 140$ was fitted to cortical neurons
- a = speed of u, b = coupling, c/d = reset

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N5 Izhikevich neuron: hardware



NEW: two coupled registers — fast v (quadratic) and slow u (recovery); a, b, c, d set the cell type

- two coupled registers v and u
- the 0.04 is $\times 41$ then $\gg 18$ (no divider)
- dt = 0.5 ms per clock, as in the original code

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N5 Izhikevich neuron: Verilog

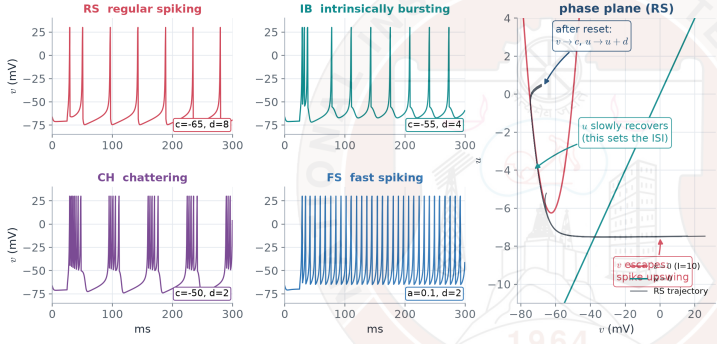
```
    localparam integer VPEAK = 30*256;
    // 0.04 v^2 in Q8: (v*v)/6400 ; implemented as (v*v*41) >>> 18 (41/2^18 = 1/6394)
    wire signed [63:0] vsq   = v * v;
    wire signed [63:0] quad  = (vsq * 41) >>> 18;
    wire signed [63:0] dv    = quad + 5*v + 140*256 - u + i_in;
    wire signed [63:0] bv    = (B * v) >>> 8;
    wire signed [63:0] du    = (A * (bv - u)) >>> 8;
    wire signed [31:0] v_next = v + (dv >>> DT_SHIFT);
    wire signed [31:0] u_next = u + (du >>> DT_SHIFT);
    always @(posedge clk) begin
        if (rst) begin v <= -65*256; u <= (B * (-65*256)) >>> 8; spike <= 1'b0; end
        else if (en) begin
            if (v_next >= VPEAK) begin
                v <= C; u <= u_next + D; spike <= 1'b1;
            end else begin
                v <= v_next; u <= u_next; spike <= 1'b0;
            end
        end else spike <= 1'b0;
    end
endmodule
```

rtl/neurons/izh_neuron.v

- quad = (v*v*41) »> 18 is 0.04 v^2
- parameters A, B, C, D pick the cell type
- reset: v ← c, u ← u + d

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N5 Izhikevich neuron: neural dynamics



How to read it: Four traces from one circuit. Right: the phase plane, v against u, with the two nullclines.

- the same RTL, 4 parameter sets → 4 firing patterns
- phase plane: after reset, u slowly recovers along the left branch
- that slow recovery sets the inter-spike interval

In the lab you should see: same circuit, four parameter sets, four rhythms; FS fires most, RS adapts, CH fires in bursts.

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N5 Izhikevich neuron: lab E05 on the Basys 3

Folder: `experiments/E05_izhikevich/`
Open: `vivado/E05_izhikevich.xpr`

control	what it does
`sw[4:0]`	input current I (try 10)
`sw[15:14]`	cell type: 00 RS, 01 IB, 10 CH, 11 FS
`btnC`	reset
`LED14..0`	v from -80 mV (empty) to +30 mV (full)
`LED15`	spike
`7-segment`	v in mV (with minus sign)

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- RS: evenly spaced flashes
- IB: a quick burst, then single spikes
- CH: repeated bursts
- FS: rapid, steady flashing

Simulation must print PASS with:

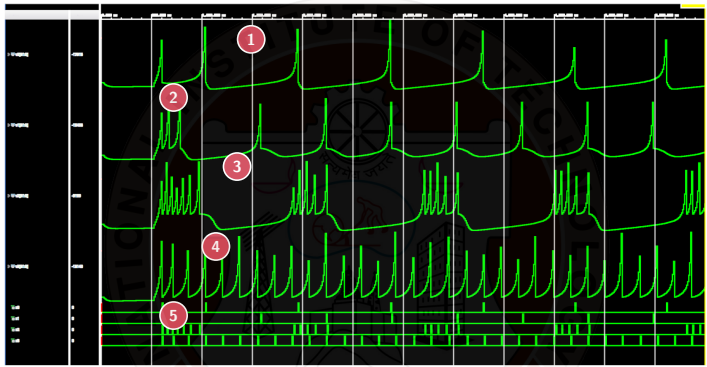
```
spikes_RS   7
spikes_IB   10
spikes_CH   25
spikes_FS   32
```

Try it: Add a fifth cell: a=0.02, b=0.25, c=-65, d=0.05 (low-threshold spiking).

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N5 Izhikevich neuron: Vivado simulation

four cell types from one circuit



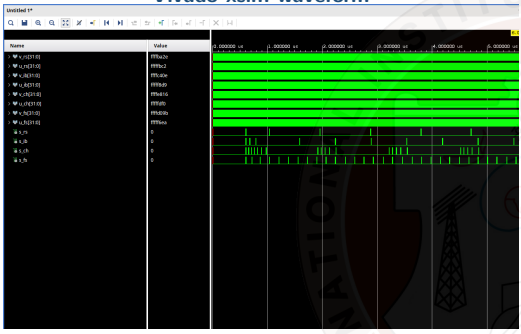
1 RS: regular, slowly spreading spikes 2 IB: a burst, then single spikes 3 CH: repeated bursts 4 FS: fast, steady spikes 5 spike outputs s0-s3

How to read it: Four rows of v (analog, in mV $\times$ 256), same input current, different a, b, c, d. The resting level is negative (about -70 mV), which is why signed decimal is needed.

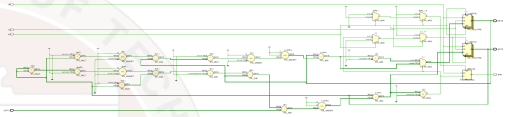
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N5 Izhikevich neuron: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic

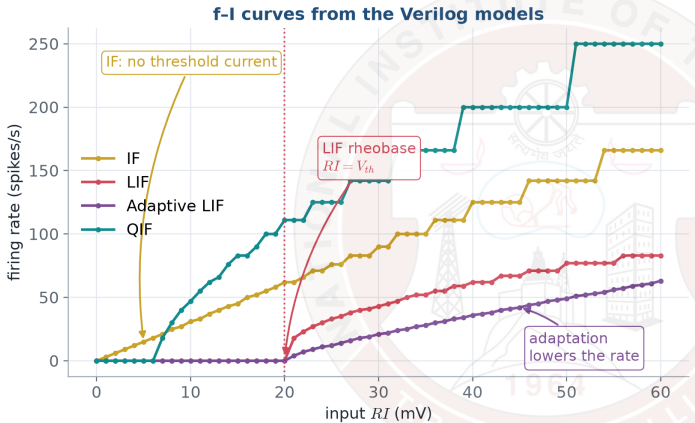


LUT	FF	DSP	F_{max}
634	65	4	50 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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Neurons compared: f–I curves



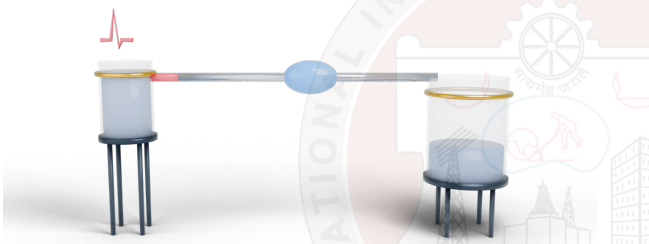
- IF fires for any input: no rheobase
- the leak creates a threshold current
- adaptation lowers the rate at the same input
- QIF: lower rheobase, steep onset

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Synapses

S1 Delta (static) synapse
S2 Exponential synapse
S3 Alpha synapse
S4 Tsodyks–Markram dynamic synapse

S1 Delta synapse



Analogy: The pipe delivers one splash

NEW: the starting point

- a synapse is the pipe from one tank's dump into the next tank
- delta synapse: each upstream dump arrives as one instant splash of size W
- nothing remains in the pipe afterwards
- the pipe width W is the synaptic weight

standard ▪ lab: `experiments/E06_delta_synapse`

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S1 Delta synapse: the equation and its physics

MODEL

$$I_{syn}(t) = \underbrace{W}_{\text{weight}} \sum_k \underbrace{\delta(t - t_k)}_{\text{pre spike times}}$$

HARDWARE (ONE CLOCK = ONE STEP)

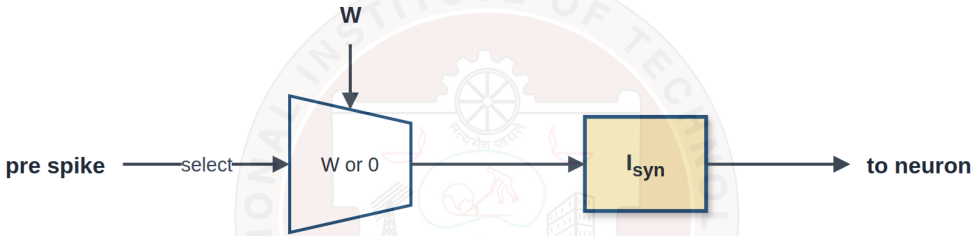
$$I_{syn}[n] = \mathsf{pre}[n] \ ? \ W : 0$$

The physics: An ideal, instantaneous charge transfer: each spike delivers a fixed packet of charge W at once, with no memory afterwards.

- the current is W at each spike time, zero otherwise

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S1 Delta synapse: hardware



no memory: the current lasts exactly one step

- just a mux: W or 0
- no state, cheapest synapse (1 LUT)

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S1 Delta synapse: Verilog

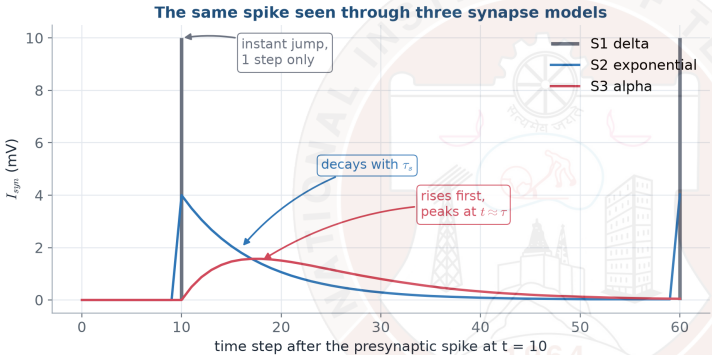
```
    always @(posedge clk) begin
        if (rst)      i_syn <= 0;
        else if (en) i_syn <= pre_spike ? W : 0;
    end
endmodule
```

rtl/synapses/delta_synapse.v

- i_syn <= pre_spike ? W : 0
- lasts exactly one clock

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S1 Delta synapse: neural dynamics



How to read it: One spike at t = 10 seen through three synapse models; compare the shapes, not only the heights.

- a one-step pulse (grey bar)
- compare with S2 and S3 on the same spike

In the lab you should see: a one-step pulse of height W at every spike, zero otherwise.

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S1 Delta synapse: lab E06 on the Basys 3

Folder: `experiments/E06_delta_synapse/`
Open: `vivado/E06_delta_synapse.xpr`

control	what it does
`btnU`	one presynaptic spike
`sw[0]`, `sw[3:1]`	random spike train on/off, rate
`btnC`	reset
`LED14..0`	synaptic current (full = 10 mV)
`LED15`	presynaptic spike
`7-segment`	I_syn in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- press btnU: the bar jumps full for one step only, then empty

Simulation must print PASS with:

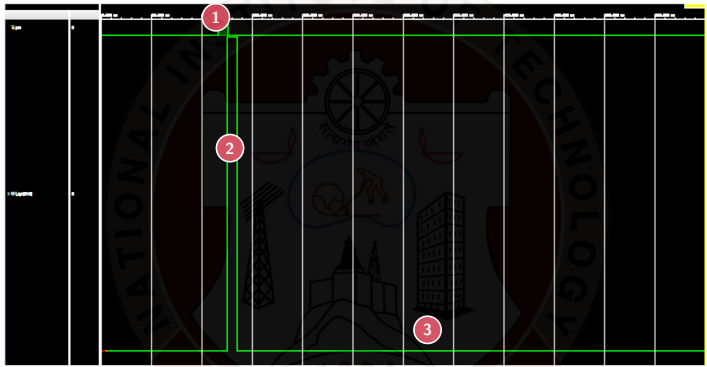
```
peak_single_Q8    2560
peak_burst_Q8     2560
```

Try it: Change W. What is the largest current this synapse can deliver?

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S1 Delta synapse: Vivado simulation

one spike



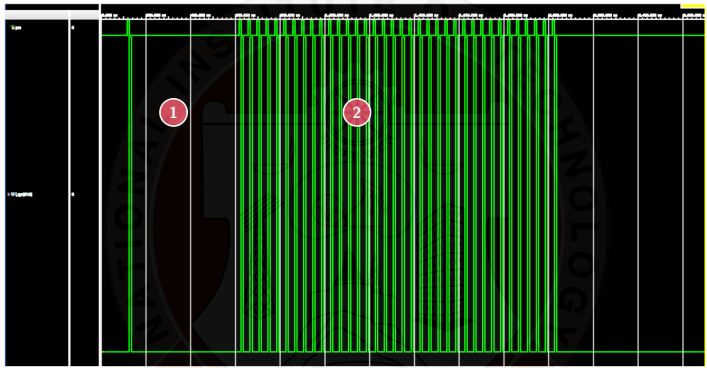
1 pre spike 2 full current for exactly one step 3 zero afterwards: no memory

How to read it: The delta synapse turns a spike into a single one-step packet of current W, then forgets it.

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S1 Delta synapse: Vivado simulation

a spike burst



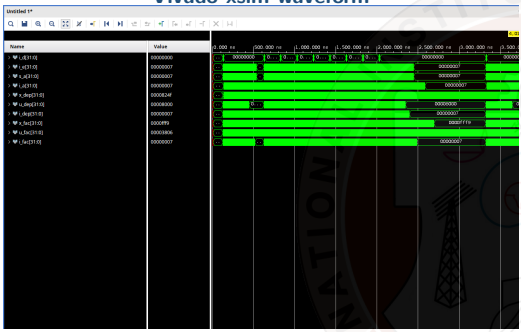
1 single spike 2 every spike gives the same height: no summation

How to read it: Even when spikes come fast, the current never builds up, because nothing is remembered between steps.

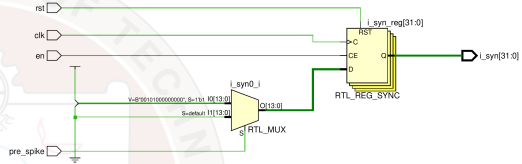
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S1 Delta synapse: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



LUT	FF	DSP	F_{max}
1	1	0	676 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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S2 Exponential synapse



Analogy: A funnel in the pipe

NEW: the flow lingers (decay)

- the splash first lands in a funnel with a small outlet
- the downstream tank gets a strong flow at once, then a fading trickle (τ_s)
- dumps that arrive close together pile up in the funnel: they add up
- same law as the tank drain: outflow $\propto$ how full the funnel is

Dayan & Abbott 2001 ▪ lab: `experiments/E07_exp_synapse`

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S2 Exponential synapse: the equation and its physics

MODEL

$$\underbrace{\tau_s \frac{dI}{dt}}_{\tau_s\text{: PSC time}} = \underbrace{-I}_{\text{decay}} + W \sum_k \delta(t - t_k)$$

HARDWARE (ONE CLOCK = ONE STEP)

$$I[n+1] = I[n] - \underbrace{I[n] \gg k}_{\text{same leak as LIF}} + W \cdot \text{pre}[n]$$

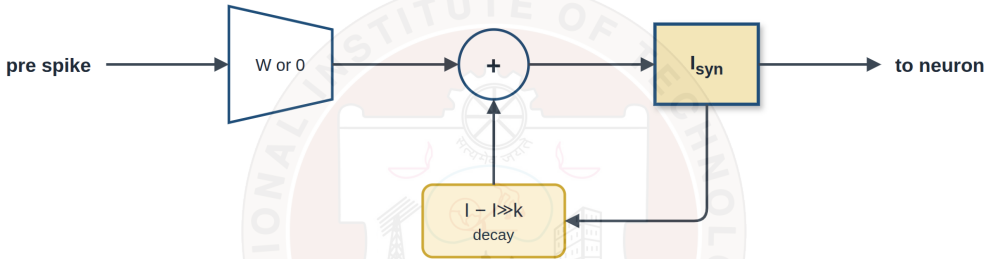
single spike: $I(t) = We^{-t/\tau_s}$

The physics: A funnel or RC discharge: the outflow is proportional to what is left (dI/dt = –I/τ_s), so the current falls by the same fraction every step: an exponential. After one τ only 37% remains. Spikes that arrive close together add up (temporal summation).

- the same first-order leak as LIF, applied to the current
- single spike: W·exp(–t/τ_s)

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S2 Exponential synapse: hardware



NEW: memory — the same leak circuit as LIF, now applied to the current

- one register, one adder, one shift
- identical structure to the LIF leak

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S2 Exponential synapse: Verilog

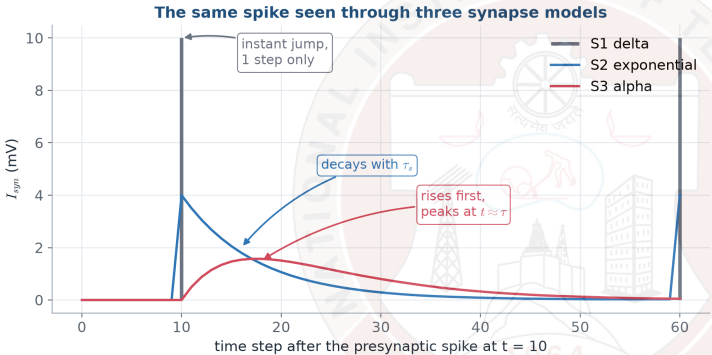
```
    always @(posedge clk) begin
        if (rst)      i_syn <= 0;
        else if (en) i_syn <= i_syn - (i_syn >>> TAU_SHIFT) + (pre_spike ? W : 0);
    end
endmodule
```

rtl/synapses/exp_synapse.v

- i_syn − (i_syn »> TAU_SHIFT) + W·pre
- TAU_SHIFT = 3 → τ_s = 8 steps

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S2 Exponential synapse: neural dynamics



How to read it: One spike at t = 10 seen through three synapse models; compare the shapes, not only the heights.

- jump of W at the spike, then exponential decay
- temporal summation in a burst

In the lab you should see: jump of W at each spike, then exponential decay; peaks build up inside a burst.

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S2 Exponential synapse: lab E07 on the Basys 3

Folder: `experiments/E07_exp_synapse/`
Open: `vivado/E07_exp_synapse.xpr`

control	what it does
`btnU`	one presynaptic spike
`sw[0], sw[3:1]`	random spike train on/off, rate
`btnC`	reset
`LED14..0`	synaptic current (full = 10 mV)
`LED15`	presynaptic spike
`7-segment`	I_syn in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- one press: the bar jumps, then shrinks by the same fraction each step
- press quickly several times: the bar climbs higher (summation)

Simulation must print PASS with:

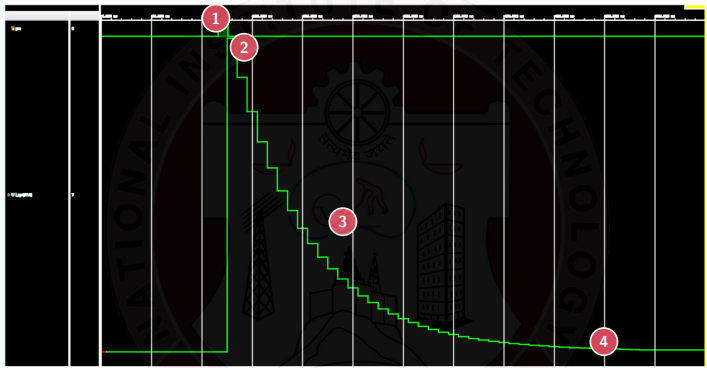
```
peak_single_Q8    1024
peak_burst_Q8     2477
```

Try it: Measure how many steps the current needs to fall to 37% of its peak. Compare with $\tau_s = 2\hat{}$TAU_SHIFT.

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S2 Exponential synapse: Vivado simulation

one spike



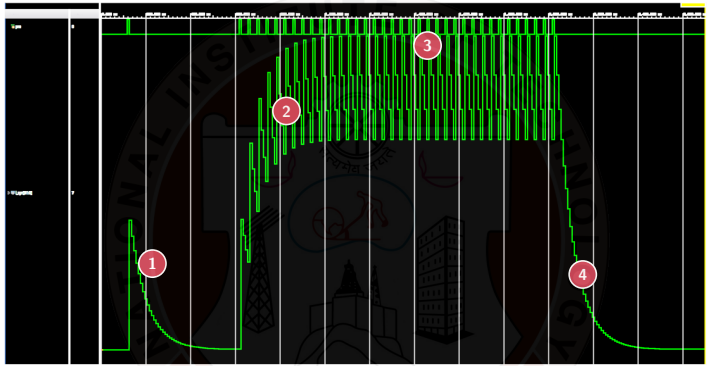
1 pre spike 2 jump of W 3 the same fraction is lost every step 4 almost 0 after about 3τ

How to read it: This staircase is the exponential: each step removes 1/8 ($\tau_s = 8$ steps), so it drops fast at first and slowly later, like a funnel emptying.

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S2 Exponential synapse: Vivado simulation

a spike burst



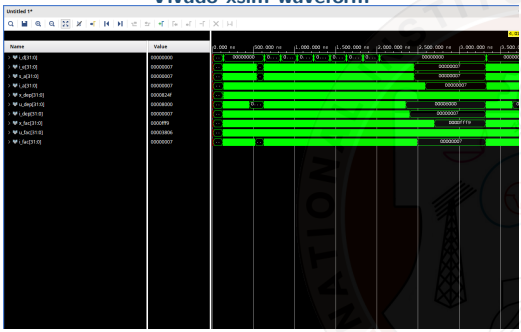
1 single spike: decay 2 spikes pile up: summation 3 plateau: input balances the decay 4 burst ends: exponential decay

How to read it: Spikes that arrive before the funnel is empty add on top: the current climbs until each spike just replaces what drained.

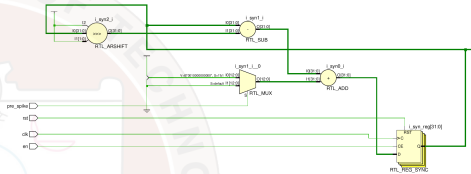
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S2 Exponential synapse: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



LUT	FF	DSP	F_{max}
51	32	0	354 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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S3 Alpha synapse



Analogy: Two funnels in series

NEW: a rise time

- the first funnel empties into a second funnel, which feeds the tank
- the flow into the tank starts at zero, rises, peaks, then fades
- the current builds up smoothly instead of jumping
- this is closer to a real synaptic current (Rall 1967)

Rall 1967 ▪ lab: `experiments/E08_alpha_synapse`

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S3 Alpha synapse: the equation and its physics

MODEL

$$\tau \frac{ds}{dt} = -s + W \sum_k \delta(t - t_k), \qquad \tau \frac{dI}{dt} = -I + \underbrace{s}_{\text{input from stage 1}}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$s \mathrel{+}= -(s \gg k) + W\,\text{pre}, \qquad I \mathrel{+}= -(I \gg k) + (s \gg k)$$

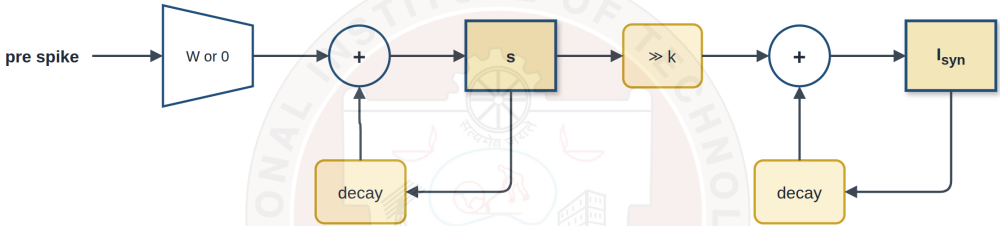
single spike: $I(t) \propto \frac{t}{\tau} e^{1-t/\tau}$ peak at $t = \tau$ (Rall 1967)

The physics: Two funnels (two RC stages) in series: the first fills instantly and drains into the second, so the current into the neuron first rises, peaks at t = τ, then decays. Real synaptic currents look like this.

- two first-order stages in series
- peak at t = τ, shape (t/τ)·exp(1 − t/τ)

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S3 Alpha synapse: hardware



NEW: two exponential stages in series — stage 1 makes the rise, stage 2 the decay

- two exponential stages chained
- stage 1 = rise, stage 2 = decay

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S3 Alpha synapse: Verilog

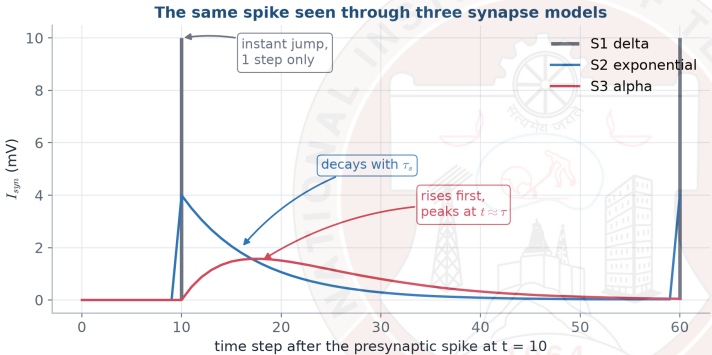
```
    always @(posedge clk) begin
        if (rst) begin s <= 0; i_syn <= 0; end
        else if (en) begin
            s     <= s - (s >>> TAU_SHIFT) + (pre_spike ? W : 0);
            i_syn <= i_syn - (i_syn >>> TAU_SHIFT) + (s >>> TAU_SHIFT);
        end
    end
endmodule
```

rtl/synapses/alpha_synapse.v

- s collects the spike, i_syn follows s
- both use the same shift (equal τ)

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S3 Alpha synapse: neural dynamics



How to read it: One spike at t = 10 seen through three synapse models; compare the shapes, not only the heights.

- smooth rise, peak, slow decay
- smaller peak than S2 but the same area

In the lab you should see: no jump: the current rises, peaks about τ steps after the spike, then decays.

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S3 Alpha synapse: lab E08 on the Basys 3

Folder: `experiments/E08_alpha_synapse/`
Open: `vivado/E08_alpha_synapse.xpr`

control	what it does
`btnU`	one presynaptic spike
`sw[0], sw[3:1]`	random spike train on/off, rate
`btnC`	reset
`LED14..0`	synaptic current (full = 5 mV)
`LED15`	presynaptic spike
`7-segment`	I_syn in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- one press: the bar grows for a few steps, then shrinks: a smooth rise and fall

Simulation must print PASS with:

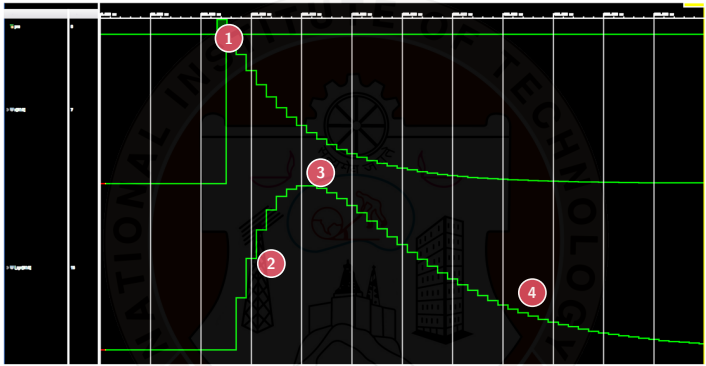
```
peak_single_Q8    402
peak_burst_Q8     2078
```

Try it: Find the step of the peak after a single spike and compare with τ.

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S3 Alpha synapse: Vivado simulation

one spike



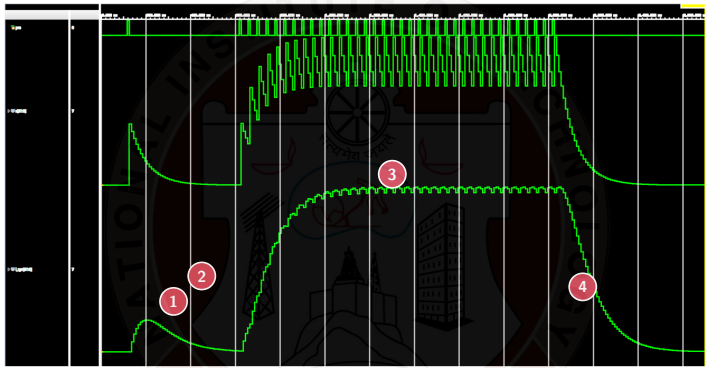
1 stage 1 (s) jumps, then drains 2 stage 2 (I) rises smoothly 3 peak about τ after the spike 4 then decays

How to read it: Two funnels in series: the second one fills while the first empties into it, so the current rises, peaks and falls instead of jumping.

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S3 Alpha synapse: Vivado simulation

a spike burst



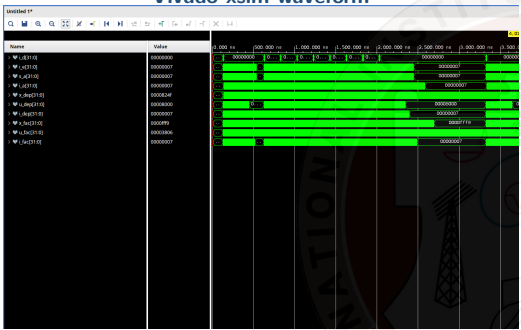
1 single spike: small smooth bump 2 smooth rise 3 smooth plateau 4 smooth decay

How to read it: Top: stage 1 (s) has the sawtooth. Bottom: the current that reaches the neuron is smooth, which is what real synaptic currents look like.

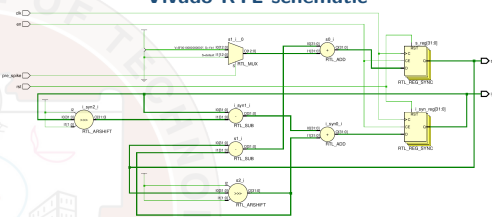
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S3 Alpha synapse: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic

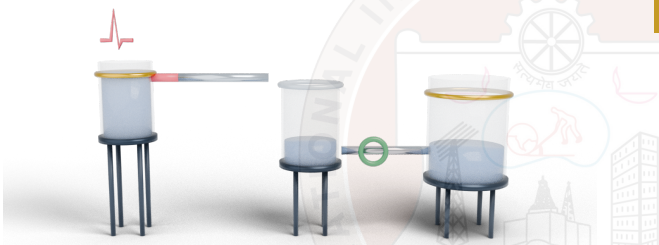


LUT	FF	DSP	F_{max}
112	64	0	246 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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S4 Tsodyks–Markram synapse



Analogy: A storage tank on the pipe that runs low

NEW: use-dependent pipe

- the pipe has its own storage tank x; each dump releases a fraction u of it
- depression: fast dumps empty the storage tank faster than it refills
- facilitation: the release valve opens wider with each use (u grows), then relaxes
- after a pause the storage refills and the valve relaxes: both recover

Tsodyks & Markram 1997 ▪ lab: `experiments/E09_tsodyks_markram`

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S4 Tsodyks–Markram synapse: the equation and its physics

MODEL

$$\frac{dx}{dt} = \frac{1-x}{\underbrace{\tau_{rec}}_{\text{resources recover}}}, \qquad \frac{du}{dt} = \frac{U-u}{\underbrace{\tau_{fac}}_{\text{utilisation relaxes}}}$$

HARDWARE (ONE CLOCK = ONE STEP)

on spike: $u \leftarrow u + U(1-u)$, $\underbrace{r = u\,x}_{\text{released fraction}}$, $x \leftarrow x - r$, $I \mathrel{+}= Wr$

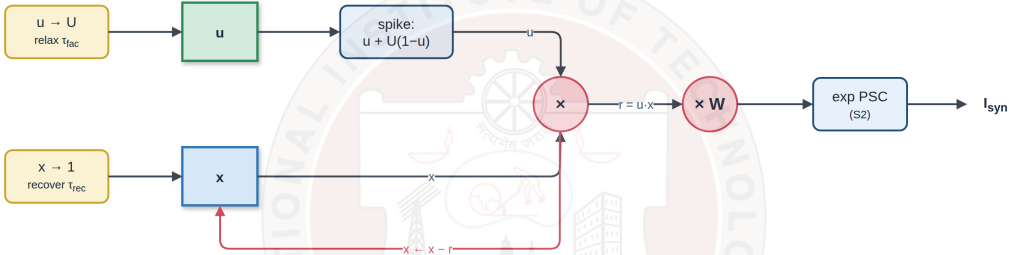
depression: high U, long τ_{rec} **facilitation**: low U, long τ_{fac} (Tsodyks & Markram 1997)

The physics: A storage tank on the pipe: each spike releases a fraction u of the available resource x, and x refills exponentially (τrec). Fast spiking drains it (depression); a growing release fraction u gives facilitation. Both recover with time.

- x = available resources, u = release probability
- each spike releases r = u·x

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S4 Tsodyks–Markram synapse: hardware



NEW: two more state registers make the effective weight depend on recent spikes

- two state registers x and u
- two multipliers (u·x and W·r)

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S4 Tsodyks–Markram synapse: Verilog

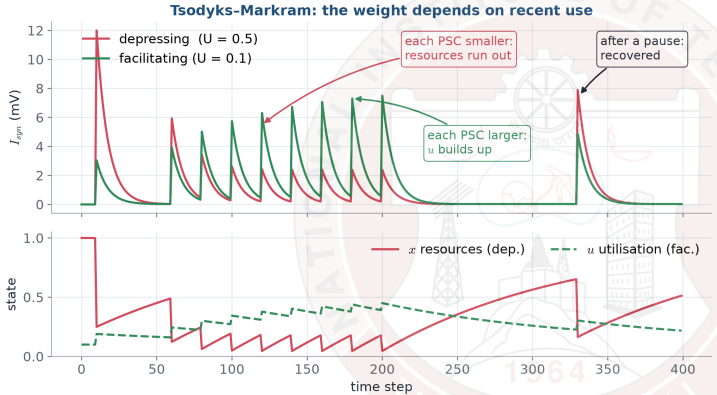
```
    wire signed [31:0] x_rec = x + ((ONE - x) >>> TAU_REC);
    wire signed [31:0] u_rel = u + ((U - u) >>> TAU_FAC);
    wire signed [63:0] u_jmp = u_rel + ((U * (ONE - u_rel)) >>> 16);
    wire signed [63:0] r     = (u_jmp * x_rec) >>> 16;          // Q16
    wire signed [63:0] i_add = (W * r) >>> 16;                  // Q8
    always @(posedge clk) begin
        if (rst) begin x <= ONE; u <= U; i_syn <= 0; end
        else if (en) begin
            if (pre_spike) begin
                u     <= u_jmp;
                x     <= x_rec - r;
                i_syn <= i_syn - (i_syn >>> TAU_S) + i_add;
            end else begin
                u     <= u_rel;
                x     <= x_rec;
                i_syn <= i_syn - (i_syn >>> TAU_S);
            end
        end
    end
endmodule
```

rtl/synapses/tm_synapse.v

- u_jmp $= u + U(1-u)$ on a spike
- $r = u_jmp \cdot x$, then $x \leftarrow x - r$
- the PSC is the exponential synapse (S2)

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S4 Tsodyks–Markram synapse: neural dynamics



How to read it: Top: the current for a spike train. Bottom: the hidden states x (resources) and u (release fraction).

- depressing: PSCs shrink 3072 → 612 over a burst
- facilitating: PSCs grow 778 → 1919
- after the pause, both recover

In the lab you should see: depressing: the last burst peak is much smaller than the first; facilitating: much larger.

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S4 Tsodyks–Markram synapse: lab E09 on the Basys 3

Folder: `experiments/E09_tsodyks_markram/`
Open: `vivado/E09_tsodyks_markram.xpr`

control	what it does
`btnU`	one presynaptic spike
`sw[0]`, `sw[3:1]`	regular spike train on/off, period
`sw[15]`	0 = depressing synapse, 1 = facilitating
`btnC`	reset
`LED14..0`	synaptic current (full = 12 mV)
`LED15`	presynaptic spike
`7-segment`	I_syn in mV

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw[15] = 0, train on: the bar gets shorter spike after spike (depression)
- sw[15] = 1: the bar gets taller spike after spike (facilitation)
- train off, wait, press btnU: the full response is back (recovery)

Simulation must print PASS with:

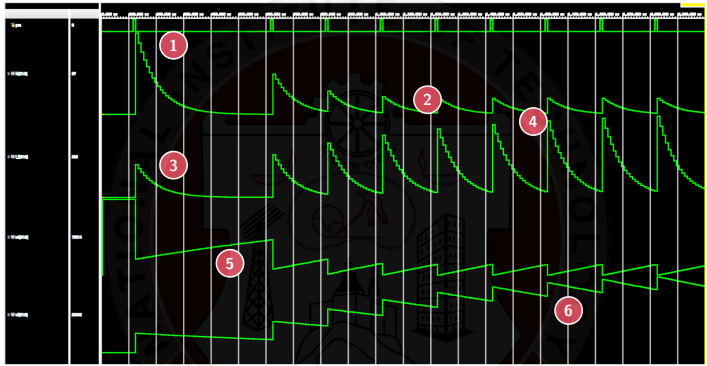
`dep_first_peak_Q8`	1521
`dep_last_peak_Q8`	612
`fac_first_peak_Q8`	1007
`fac_last_peak_Q8`	1919

Try it: Shorten the train period. Does depression get stronger?

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S4 Tsodyks–Markram synapse: Vivado simulation

depression and facilitation



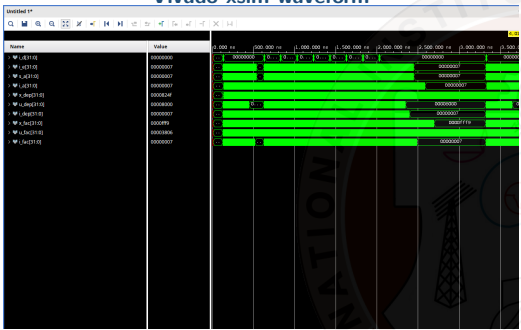
1 depressing: first response large 2 later responses smaller 3 facilitating: first response small 4 grows with every spike 5 resources x: drop and refill 6 release fraction u builds up

How to read it: Same spike train into two synapses. Rows 2 and 3 are the currents; rows 4 and 5 are the hidden states that cause the difference.

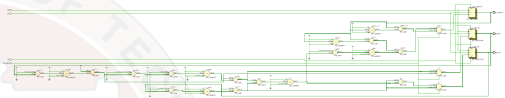
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S4 Tsodyks–Markram synapse: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



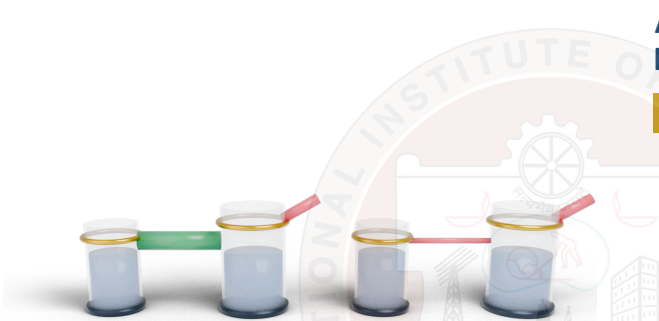
LUT	FF	DSP	F_{max}
435	96	6	49 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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Learning

L1 Spike-timing-dependent plasticity (STDP)
L2 STDP network: unsupervised learning



Analogy: Pipes widen if they helped

NEW: learning: pipe width changes

- water from A arrived just before tank B dumped: the pipe helped, so widen it (LTP)
- water arrived just after B dumped: it was wasted, so narrow the pipe (LTD)
- each tank leaves a wet mark that dries out: traces remember how recent a dump was
- the shorter the gap, the bigger the change

Bi & Poo 1998; Song et al. 2000 ▪ lab: `experiments/E10_stdp_synapse`

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L1 STDP: the equation and its physics

MODEL

$$\Delta w = \begin{cases} A_+ \, e^{-\Delta t/\tau_+} & \Delta t > 0 \text{ (pre before post: LTP)} \\ -A_- \, e^{\Delta t/\tau_-} & \Delta t < 0 \text{ (post before pre: LTD)} \end{cases}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$\underbrace{x_{pre} \mathrel{+}= A_+}_{\text{on pre spike}}, \quad \underbrace{x_{post} \mathrel{+}= A_-}_{\text{on post spike}}; \quad \text{post: } w \mathrel{+}= x_{pre}, \quad \text{pre: } w \mathrel{-}= x_{post}$$

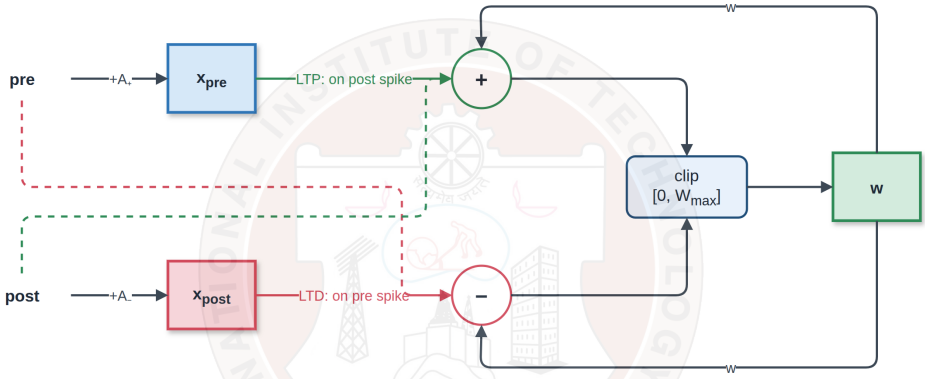
traces decay like the exponential synapse; w clipped to $[0, W_{max}]$ (Bi & Poo 1998; Song et al. 2000)

The physics: Each neuron leaves a decaying trace (an exponential memory, like a wet mark drying). When the post neuron fires while the pre trace is still high, the weight grows (cause before effect); the reverse order shrinks it.

- $\Delta t > 0$: potentiation (LTP), $\Delta t < 0$: depression (LTD)
- implemented with two decaying traces, no timestamps

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L1 STDP: hardware



NEW: learning — each trace remembers how recently its neuron fired; the other side's spike reads it

- x_pre remembers recent pre spikes, x_post recent post spikes
- a post spike reads x_pre (LTP), a pre spike reads x_post (LTD)

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L1 STDP: Verilog

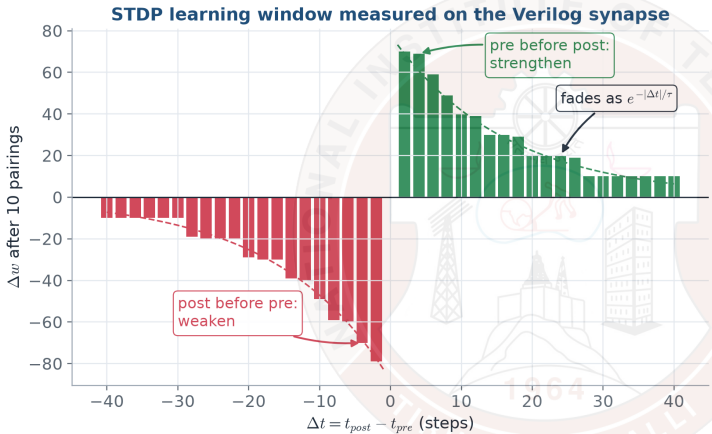
```
    wire signed [31:0] xpre_d  = x_pre  - (x_pre  >>> TAU_PLUS);
    wire signed [31:0] xpost_d = x_post - (x_post >>> TAU_MINUS);
    wire signed [31:0] ltp = post_spike ? (xpre_d  >>> TR_FRAC) : 0;
    wire signed [31:0] ltd = pre_spike  ? (xpost_d >>> TR_FRAC) : 0;
    wire signed [31:0] w_new = w + ltp - ltd;
    always @(posedge clk) begin
        if (rst) begin w <= W_INIT; x_pre <= 0; x_post <= 0; end
        else if (en) begin
            x_pre  <= xpre_d  + (pre_spike  ? (A_PLUS  << TR_FRAC) : 0);
            x_post <= xpost_d + (post_spike ? (A_MINUS << TR_FRAC) : 0);
            if (learn)
                w <= (w_new < 0) ? 0 : (w_new > WMAX) ? WMAX : w_new;
        end
    end
endmodule
```

rtl/learning/stdp_synapse.v

- traces decay like the exponential synapse
- ltp/ltd gated by the opposite spike
- w is clipped to [0, WMAX]

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L1 STDP: neural dynamics



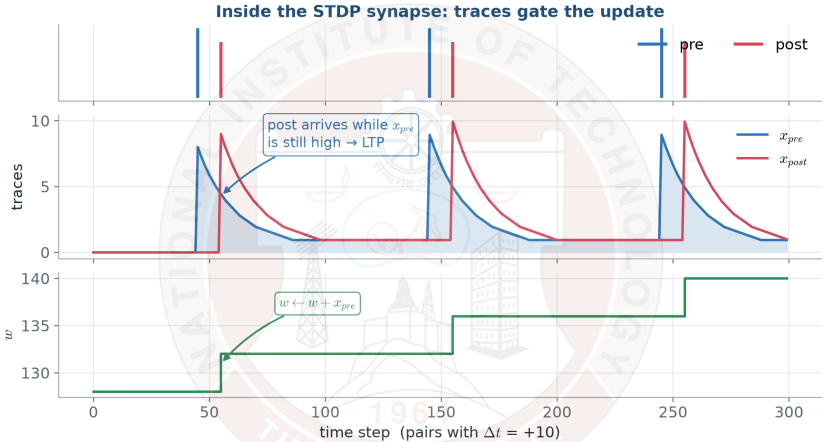
How to read it: Each bar is the weight change after 10 pairings at one timing Δt; green = stronger, red = weaker.

- measured window: exponential on both sides
- LTD slightly larger than LTP ($A_- > A_+$)
- trace view: post arrives while x_pre is high → w steps up

In the lab you should see: $\Delta t = +10$ (pre first): weight increases; $\Delta t = -10$ (post first): weight decreases; the window decays with $|\Delta t|$.

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L1 STDP: inside the synapse



How to read it: the blue trace jumps at each pre spike and fades; when the post spike arrives (red) the weight steps up by the trace's current height. Closer spikes, bigger step.

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L1 STDP: lab E10 on the Basys 3

Folder: `experiments/E10_stdp_synapse/`
Open: `vivado/E10_stdp_synapse.xpr`

control	what it does
`btnL`	presynaptic spike
`btnR`	postsynaptic spike
`btnC`	reset (w back to 128)
`LED14..0`	synaptic weight w (full = 255)
`LED15`	any spike
`7-segment`	weight w

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- press btnL, then quickly btnR: w goes UP (pre before post, LTP)
- press btnR, then quickly btnL: w goes DOWN (LTD)
- press them far apart in time: w barely changes (traces have dried out)

Simulation must print PASS with:

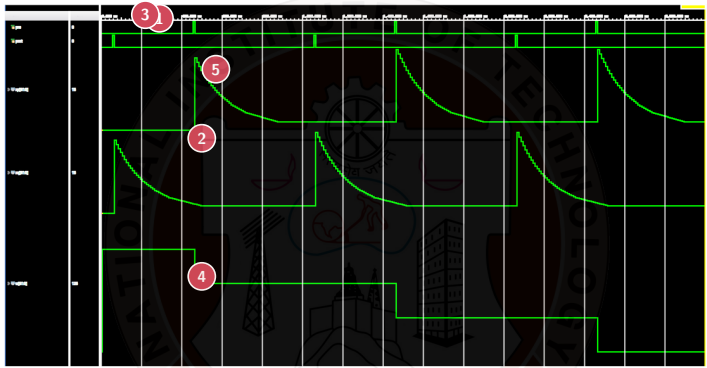
```
dw_at_plus10    40
dw_at_minus10   -49
```

Try it: Plot stdp_window.csv: dw against Δt (the learning window).

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L1 STDP: Vivado simulation

post before pre (Δt = -40)



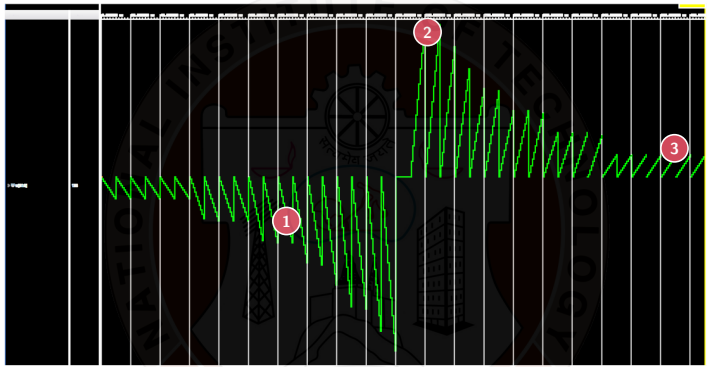
1 post fires first 2 post trace starts to decay 3 pre arrives later 4 w drops by the remaining post trace (LTD) 5 pre trace

How to read it: Each spike leaves a decaying trace. When pre fires, the weight is reduced by whatever is left of the post trace: post before pre weakens the synapse.

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L1 STDP: Vivado simulation

the whole timing sweep



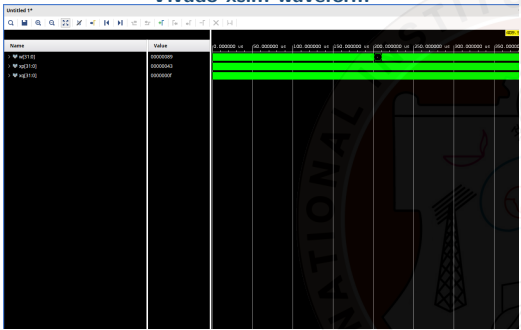
1 $\Delta t < 0$: weight falls (LTD) 2 $\Delta t > 0$: weight grows (LTP) 3 large $|\Delta t|$: small change

How to read it: w is reset to 128 for each timing Δt (from -40 to +40) and paired 10 times: the depth of each tooth is the learning window.

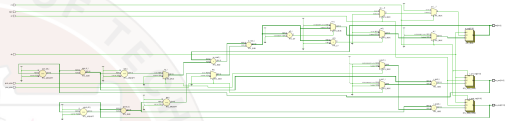
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L1 STDP: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic

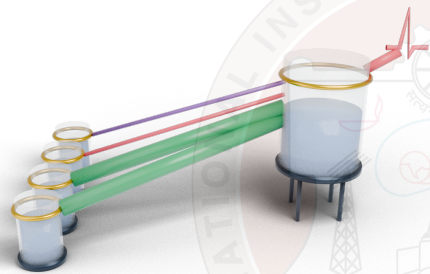


LUT	FF	DSP	F_{max}
185	95	0	134 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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L2 STDP network



Analogy: Four pipes into one tank

NEW: learning a pattern

- four upstream tanks, all dumping equally often, feed one downstream tank
- two of them always dump together; their joint splash makes the tank dump
- so their pipes keep widening while the random pipes narrow
- nobody told the system which pair matters: unsupervised learning

Song, Miller & Abbott 2000 ▪ lab: `experiments/E11_stdp_network`

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L2 STDP network: the equation and its physics

MODEL

$$\tau_s \frac{dI}{dt} = -I + \sum_{i=0}^{3} \underbrace{w_i}_{\text{learned by STDP}} \cdot \text{GAIN} \cdot \underbrace{s_i(t)}_{\text{input spikes}}$$

HARDWARE (ONE CLOCK = ONE STEP)

$$I \mathrel{+}= -(I \gg 2) + \Sigma_i \, \text{pre}_i \, w_i \, \text{GAIN} \;\rightarrow\; \text{LIF} \;\rightarrow\; \text{post}$$

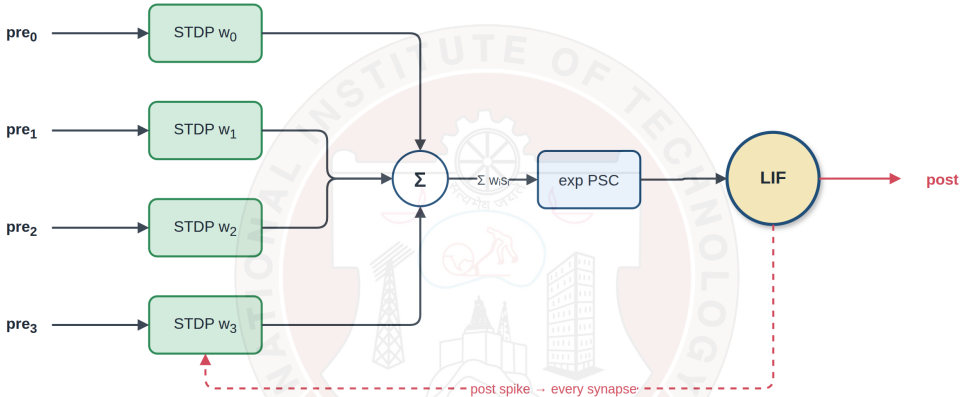
stable competition needs $A_- > A_+$ (here 13 vs 8), as in Song, Miller & Abbott 2000

The physics: Competition through timing: the inputs that fire together make the output neuron fire, so their synapses keep getting the 'cause' credit and grow; random inputs mostly arrive at the wrong time and shrink.

- the neuron sums w·spike over its inputs
- competition is stable only if $A_- > A_+$

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L2 STDP network: hardware



- reuse: $4 \times$ stdp_synapse $+ 1 \times$ lif_neuron
- the post spike is fed back to every synapse

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L2 STDP network: Verilog

```
    always @(*) begin
        drive = 0;
        for (k = 0; k < N_IN; k = k + 1)
            if (pre[k]) drive = drive + $signed(w_bus[32*k +: 32]) * GAIN;
    end
    always @(posedge clk) begin
        if (rst)      i_syn <= 0;
        else if (en) i_syn <= i_syn - (i_syn >>> 2) + drive;
    end

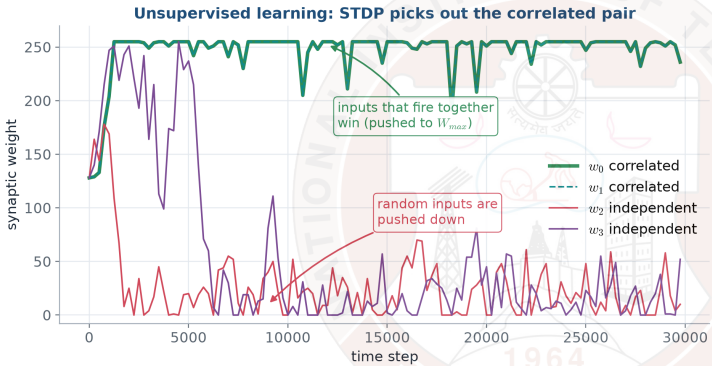
    lif_neuron #(.TAU_SHIFT(3), .T_REF(3)) u_lif (
        .clk(clk), .rst(rst), .en(en), .i_in(i_syn), .v(v), .spike(post));
endmodule
```

rtl/learning/stdp_network.v

- drive = Σ w_i · GAIN over active inputs
- exponential current into the LIF neuron
- generate loop instantiates the synapses

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L2 STDP network: neural dynamics



How to read it: The four weights over 30 000 steps of random input: watch which ones climb and which ones fall.

- correlated weights go to WMAX within 2000 steps
- independent weights are pushed near zero
- the neuron has learned which inputs belong together

In the lab you should see: correlated inputs 0 and 1 end near 255, independent inputs 2 and 3 near 0.

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L2 STDP network: lab E11 on the Basys 3

Folder: `experiments/E11_stdp_network/`
Open: `vivado/E11_stdp_network.xpr`

control	what it does
`sw[0]`	inputs on
`sw[1]`	inputs 0 and 1 correlated (1) or independent (0)
`sw[2]`	learning on
`sw[7:4]`	input rate (try 0001)
`sw[15:14]`	which weight w0..w3 to show
`btnC`	reset all weights to 128
`LED14..0`	selected weight (full = 255)
`LED15`	output neuron spike
`7-segment`	selected weight

1. Run Behavioral Simulation (expect PASS)
2. Generate Bitstream
3. Program Device, then try:

Try on the board:

- sw[0]=1, sw[1]=1, sw[2]=1, sw[7:4]=0001: after 30 s, w0 and w1 near 255, w2 and w3 near 0
- repeat with sw[1]=0: no pair is special, so no clear winner

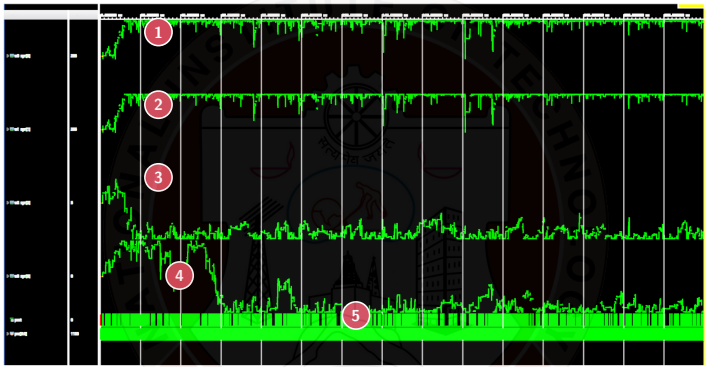
Simulation must print PASS with:
`(behavioural check)` see expected behaviour

Try it: Correlate inputs 2 and 3 instead. Which pair wins now?

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L2 STDP network: Vivado simulation

learning in a 4-input network



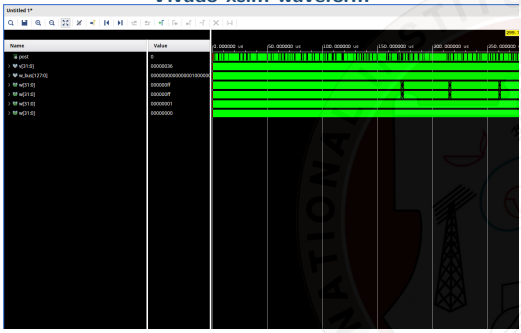
1 w0 rises to 255 2 w1 rises to 255 3 w2 falls 4 w3 falls 5 output spikes

How to read it: All four weights start at 128 on the same 0-256 scale. Inputs 0 and 1 fire together, so they win; the random inputs are pushed down. No teacher, only timing.

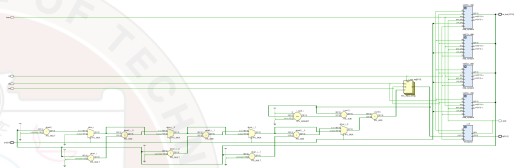
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L2 STDP network: what Vivado shows

Vivado xsim waveform



Vivado RTL schematic



LUT	FF	DSP	F_{max}
1164	455	0	112 MHz

model only, Artix-7 synthesis estimate (the Basys 3 board design adds the display and step clock)

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What it costs in hardware

	Module	LUT	FF	DSP	F_{max} (MHz)
N1	Integrate-and-Fire (IF)	96	33	0	161
N2	Leaky Integrate-and-Fire (LIF)	142	41	0	132
N3	Adaptive LIF	206	65	0	112
N4	Quadratic Integrate-and-Fire (QIF)	182	33	4	57
N5	Izhikevich neuron	634	65	4	50
S1	Delta (static) synapse	1	1	0	676
S2	Exponential synapse	51	32	0	354
S3	Alpha synapse	112	64	0	246
S4	Tsodyks–Markram dynamic synapse	435	96	6	49
L1	Spike-timing-dependent plasticity (STDP)	185	95	0	134
L2	STDP network: unsupervised learning	1164	455	0	112

Vivado 2025.2, xc7a100tcsg324-1, out-of-context synthesis. Models with multipliers (QIF, Izhikevich, TM) miss 100 MHz: pipelining is the next exercise.

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Take-aways

- a spiking neuron is a small state machine: add, leak (shift), compare, reset
- the exponential is everywhere: one shift per step gives leak, synaptic decay and STDP traces
- each richer model adds one register or one multiplier, and you can feel it on the board
- STDP learns from spike timing alone, with no teacher: watch two weights win on the LEDs
- all eleven labs are ready Vivado projects for the Basys 3: simulate, build, play

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Thank you

Questions and hands-on time

D. Mohan Reddy, PMRF ▪ Guide: Dr. G. Lakshminarayanan, Professor (HAG)
Department of ECE, NIT Tiruchirappalli

Declaration: This workshop is conducted **free of cost** as part of the **national service** commitment of the Prime Minister's Research Fellowship (PMRF), through external teaching assistantship, to disseminate knowledge to neighbouring colleges.